

# Pre Survey - How Generative AI Visualizations relate to Human Mental Images

## Survey Flow

Standard: Intro & Consent (1 Question)

Standard: Consent Statement: (1 Question)

Branch: New Branch

If

If Consent Statement: By clicking “I agree”, you confirm that: You are at least 18 years old You ha... No, I do not wish to participate Is Selected

**EndSurvey: Advanced**

Standard: Screening 1 (5 Questions)

Branch: New Branch

If

If If you don't use GenAI for image generation, what do you typically use GenAI for? q://QID40/SelectedChoicesCount Is Not Empty

**EndSurvey: Advanced**

Standard: Screening 2 (3 Questions)

Standard: Tools and Access (2 Questions)

Standard: Usage Contexts and Goals (3 Questions)

Standard: Prompting Practices (6 Questions)

Standard: Perceptions (4 Questions)

Standard: Open-Ended Questions (4 Questions)

Standard: Demographics (2 Questions)

Standard: Closing (2 Questions)

Page Break

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## Start of Block: Intro & Consent

### **Purpose of the Study**

You are invited to take part in a research study examining how people visualize concepts in their mind and how generative AI (GenAI) tools (e.g., ChatGPT image generation) may help or influence this process. GenAI (Generative Artificial Intelligence) refers to a type of artificial intelligence that can create new content (text, music, videos, images, etc.), rather than only analyzing or classifying existing data. This survey investigates how AI-generated tools are used and for which purposes. The study consists of approximately 30 short questions and will take approximately **6 minutes** to complete. **Your Rights:** Participation is voluntary. You may withdraw at any time without penalty. You may skip any question you do not wish to answer. No identifying information will be collected. **Confidentiality:** Your responses will be stored securely and analyzed anonymously. No IP addresses, names, or identifying information will be linked to your data. **Contact Information:** If you have questions about the study, please contact: Gwen Ghittini – [gwen.ghittini@student.unisg.ch](mailto:gwen.ghittini@student.unisg.ch)

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Page Break

End of Block: Intro & Consent

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Start of Block: Consent Statement:

**Q1 Consent Statement:** By clicking “I agree”, you confirm that: You are at least 18 years old  
You have read and understood the information above You voluntarily agree to  
participate in this study. Do you agree to participate in this study?

- ☐ Yes, I agree to participate (1)
- ☐ No, I do not wish to participate (2)

End of Block: Consent Statement:

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Start of Block: Screening 1

Q2 Before image generators (like DALL·E (OpenAI), Midjourney, Nano Banana etc.) were available, how did you express or visualize a mental image to communicate it to others or clarify it for yourself? *Examples of such situations could include: (1) describing a scene from a story, dream, or memory to someone else; (2) explaining a design idea to a colleague or client (e.g., how a poster, logo, room, or interface should look); or (3) planning and imagining a space (e.g., rearranging furniture, designing a room, or envisioning a building or landscape).*

- ☐ Drawing or sketching by hand (on paper or digitally) (1)
- ☐ Describing the image verbally (spoken explanation) (2)
- ☐ Writing a detailed description (e.g., notes, text messages) (3)
- ☐ Using existing images (photos, illustrations, references) (4)
- ☐ Googling (5)
- ☐ Gesturing or acting things out (6)
- ☐ Creating diagrams or schematics (7)
- ☐ Using physical objects or models (8)
- ☐ Other (please specify): (9)  
\_\_\_\_\_
- ☐ Not applicable / I did not do this (10)

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Q3 Have you ever used a generative AI (GenAI) tool for image generation?

☐ Yes (1)

☐ No (2)

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*Display this question:*

*If Q3 = No*

Q4.1 If you **don't use GenAI** for image generation, what do you typically use GenAI for?

- ☐ Writing and editing text (1)
  - ☐ Paraphrasing or rephrasing text (2)
  - ☐ Information search and Q&A (3)
  - ☐ Summarizing content (4)
  - ☐ Explaining complex concepts (5)
  - ☐ Brainstorming ideas (6)
  - ☐ Translation and language support (7)
  - ☐ Programming help (generation or explanation) (8)
  - ☐ Decision support / advice (9)
  - ☐ Creative writing or storytelling (10)
  - ☐ Other (please explain shortly) (11)
-

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*Display this question:*

*If If you don't use GenAI for image generation, what do you typically use GenAI for?  
q://QID40/SelectedChoicesCount Is Greater Than or Equal to 1*

**Q4.2 Please leave me your e-mail address:**

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*Display this question:*

*If Q3 = Yes*

Q4.3 What do you typically use GenAI for in addition to using it for image generation?

- ☐ Writing and editing text (1)
  - ☐ Paraphrasing or rephrasing text (2)
  - ☐ Information search and Q&A (3)
  - ☐ Summarizing content (4)
  - ☐ Explaining complex concepts (5)
  - ☐ Brainstorming ideas (6)
  - ☐ Translation and language support (7)
  - ☐ Programming help (generation or explanation) (8)
  - ☐ Decision support / advice (9)
  - ☐ Creative writing or storytelling (10)
  - ☐ Other (please explain shortly) (11)
- 

End of Block: Screening 1

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Start of Block: Screening 2

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Q6 How long have you been using GenAI for image generation? *(as a reference: Public adoption of Gen AI surged in 2022, when tools like ChatGPT/Midjourney/DALLE/Stable Diffusion made generative AI accessible to millions of users.)*

- ☐ Less than 1 month (1)
  - ☐ 1–6 months (2)
  - ☐ 7–12 months (3)
  - ☐ 1–2 years (4)
  - ☐ More than 2 years (5)
- 

Q7 How often do you currently use GenAI for image generation?

- ☐ Daily (1)
  - ☐ Several times per week (2)
  - ☐ Weekly (3)
  - ☐ Monthly (4)
  - ☐ Less often (5)
  - ☐ I no longer use it. If so, why? (6)
- 

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*Display this question:*

*If If you don't use GenAI for image generation, what do you typically use GenAI for?*  
*q://QID40/SelectedChoicesCount Is Equal to >1*

Q8 What is your experience with prompting GenAI in general?

|                                                                | Very<br>inexperienced<br>(1) | Inexperienced<br>(2)  | Slightly<br>inexperienced<br>(3) | Neither<br>inexperienced<br>nor<br>experienced<br>(4) | Slightly<br>experienced<br>(5) | Experienced<br>(6)    |
|----------------------------------------------------------------|------------------------------|-----------------------|----------------------------------|-------------------------------------------------------|--------------------------------|-----------------------|
| 1 = very<br>inexperienced<br>to 7 = very<br>experienced<br>(1) | <input type="radio"/>        | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>                                 | <input type="radio"/>          | <input type="radio"/> |

End of Block: Screening 2

Start of Block: Tools and Access

Q9 Which GenAI image tools have you used so far?

- ☐ ChatGPT (image generation) (1)
- ☐ Midjourney (2)
- ☐ DALL·E (3)
- ☐ Stable Diffusion (local or cloud) (4)
- ☐ Adobe Firefly (5)
- ☐ Leonardo / Playground / similar tools (6)
- ☐ Nano Banana (7)
- ☐ Other (8) \_\_\_\_\_

Q10 Do you pay for GenAI image tools?

☐

No (1)

☐

Yes, occasionally (2)

☐

Yes, subscription-based (3)

End of Block: Tools and Access

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Start of Block: Usage Contexts and Goals

Q11 For what purposes do you mainly use GenAI image tools?

☐

Personal use / hobby (1)

☐

University / studies (2)

☐

Professional work (3)

☐

Creative projects (art/design) (4)

☐

Social media content (5)

☐

Gaming / roleplay / character design (6)

☐

Ideation / brainstorming (7)

☐

Other (8) \_\_\_\_\_

Q12 What types of images do you generate most frequently?

- ☐ People / characters (1)
  - ☐ Landscapes / environments (2)
  - ☐ Products / mockups (3)
  - ☐ Logos / icons / illustrations (4)
  - ☐ Photorealistic images (5)
  - ☐ Style experiments (e.g., “in the style of...”) (6)
  - ☐ Scenes / storyboards /scenerios (7)
  - ☐ Others (please describe shortly) (8)
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Q13 How important are the following goals when generating images?

|                                             | Not a<br>priority<br>(1) | Low<br>priority<br>(2) | Somewhat<br>priority (3) | Neutral<br>(4)        | Moderate<br>priority<br>(5) | High<br>priority<br>(6) | Essential<br>priority<br>(7) |
|---------------------------------------------|--------------------------|------------------------|--------------------------|-----------------------|-----------------------------|-------------------------|------------------------------|
| Realism /<br>high visual<br>quality (1)     | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |
| Creativity /<br>novelty (2)                 | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |
| Control<br>over the<br>output (3)           | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |
| Speed /<br>efficiency<br>(4)                | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |
| Variety /<br>exploration<br>(5)             | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |
| Alignment<br>with my<br>mental<br>image (6) | <input type="radio"/>    | <input type="radio"/>  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/>   | <input type="radio"/>        |

End of Block: Usage Contexts and Goals

Start of Block: Prompting Practices

Q14 What are your main challenges when prompting?

- ☐ Knowing how to start (1)
  - ☐ Finding the right words (2)
  - ☐ Controlling details (3)
  - ☐ Maintaining consistency across images (4)
  - ☐ Unexpected or incorrect interpretations (5)
  - ☐ Time spent on multiple iterations (6)
  - ☐ Others (please describe shortly) (7)
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Q15 How do you usually create prompts?

- ☐ I write them entirely myself (1)
  - ☐ I use templates and adapt them (2)
  - ☐ I ask another AI (e.g., ChatGPT) to generate prompts (3)
  - ☐ I use prompts from online communities (4)
  - ☐ A combination of the above (5)
  - ☐ Others (please describe shortly) (6)
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Q16 When using generative AI tools (e.g., ChatGPT), do you deliberately adjust how you phrase your inputs in order to obtain better or more relevant outputs? (also referred to as prompt engineering).

- ☐ Strongly disagree (1)
  - ☐ Disagree (2)
  - ☐ Somewhat disagree (3)
  - ☐ Neither agree nor disagree (4)
  - ☐ Somewhat agree (5)
  - ☐ Agree (6)
  - ☐ Strongly agree (7)
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Q17 Which elements do you frequently include in prompts?

- ☐ Artistic or style references (1)
  - ☐ Composition (e.g., perspective, framing) (2)
  - ☐ Lighting / color (3)
  - ☐ Camera or photography terms (4)
  - ☐ Materials / textures (5)
  - ☐ Textelements to be displayed on the image (6)
  - ☐ Negative prompts (explicitly stating what should not be included) (7)
  - ☐ I rarely specify details (8)
  - ☐ Others (please describe shortly) (9)
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Q18 How difficult is it for you to phrase prompts so that the output matches your intention?

|                                                                      | Extremely<br>easy (1) | Very<br>easy (2)      | Somewhat<br>easy (3)  | Neither<br>easy nor<br>difficult<br>(4) | Somewhat<br>difficult (5) | Very<br>difficult<br>(6) | Extremely<br>difficult<br>(7) |
|----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------------------------|---------------------------|--------------------------|-------------------------------|
| 1 =<br>Extremely<br>easy ... 7<br>=<br>Extremely<br>difficult<br>(1) | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>                   | <input type="radio"/>     | <input type="radio"/>    | <input type="radio"/>         |

Q19 How many iterations do you typically need to achieve a satisfying result? (Excluding prompts used for prompt engineering purposes).

- ☐ 1 (1)
  - ☐ 2-3 (2)
  - ☐ 4-6 (3)
  - ☐ 7-10 (4)
  - ☐ More than 10 (5)
  - ☐ Hard to say (please describe shortly) (6)
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End of Block: Prompting Practices

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Start of Block: Perceptions

Q20 How do you primarily perceive GenAI in image creation?

- ☐ As a tool (1)
  - ☐ As a creative partner (2)
  - ☐ As a co-author / collaborator (3)
  - ☐ Other/ It depends (please describe shortly) (4)
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Q21 Please indicate your level of agreement:

|                                                                  | Strongly disagree<br>(1) | Disagree<br>(2)       | Slightly disagree<br>(3) | Neither agree nor disagree<br>(4) | Slightly agree<br>(5) | Agree<br>(6)          | Strongly agree<br>(7) |
|------------------------------------------------------------------|--------------------------|-----------------------|--------------------------|-----------------------------------|-----------------------|-----------------------|-----------------------|
| Generative AI helps me reach results faster. (1)                 | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Generative AI inspires ideas I would not have had otherwise. (2) | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Generative AI enhances my creativity. (3)                        | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Generative AI reduces my creative autonomy. (4)                  | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| I feel that the AI understands my intentions. (5)                | <input type="radio"/>    | <input type="radio"/> | <input type="radio"/>    | <input type="radio"/>             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Q22 Considering all your use of generative AI **during one week**, what percentage of the time do you use it alone versus together with others?

0 10 20 30 40 50 60 70 80 90 100

0% = always together with others...  
100% = always alone (1)



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Q23 (If working with others) How does collaboration typically take place?

- ☐ I don't work with others (1)
  - ☐ One person writes prompts, others evaluate (2)
  - ☐ Joint prompt writing and iteration (3)
  - ☐ Alternating prompt creation (4)
  - ☐ Joint discussion of goals and criteria (5)
  - ☐ Others (please describe shortly) (6)
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End of Block: Perceptions

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Start of Block: Open-Ended Questions

Q24 Please give me an example for a typical situation in which you use GenAI for image generation.

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Q25 Please give me an example for when an AI-generated image successfully achieved what you intended.

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Q26 When does an AI-generated image fail to achieve what you intended?

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Q27 Is there anything about generative AI image tools that you find problematic or concerning?

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End of Block: Open-Ended Questions

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Start of Block: Demographics

Q28 How old are you?

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Q29 What is your gender?

- ☐ Male (1)
- ☐ Female (2)
- ☐ Non-binary / third gender (3)
- ☐ Prefer not to say (4)

End of Block: Demographics

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Start of Block: Closing

Q30 Thank you sincerely for taking the time to participate in this study! To bring my research to the finish line, I'm running a short, self-conducted experiment in the form of an online survey (about 25 minutes). As a thank-you for taking part, you'll be entered into a draw to **win two cinema ticket vouchers** 🎟️! Your participation would genuinely make a difference and help strengthen the study's results. If you'd like to join, simply leave your email address below and I'll send you the survey link. If you're curious about the results, you're very welcome to leave your email to receive a summary of the findings once the study is complete. Would you like to participate in the follow-up study?

☐

YES! I want to fill out the second survey and get a chance to win 2 cinema tickets (please write your e-mail in the text box below) (1)

☐

I'm interested in the study findings (please write your e-mail in the text box below) (2)

☐

No thanks. (3)

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Page Break

Display this question:

*If Q30 = YES! I want to fill out the second survey and get a chance to win 2 cinema tickets (please write your e-mail in the text box below)*

*Or Q30 = I'm interested in the study findings (please write your e-mail in the text box below)*

Q32 Your Email Address

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End of Block: Closing

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